

# Jaiveer Bassi

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## Experience

### Stanford University

Sep 2026 – Present

*Stanford SSG Fellow*

*Stanford, CA*

- Conduct mentor-guided independent research by developing a core curiosity question, evaluating primary scientific literature, and formulating testable hypotheses.
- Prepare research findings for a formal presentation to Stanford mentors and peers.

### Physical Intelligence

May 2026 – Present

*AI Robot Operator*

*San Francisco, CA*

- Generate high-quality demonstration data for general-purpose robotics foundation models by teleoperating robotic arms through manipulation, sorting, and assembly tasks.
- Evaluate policy rollouts against task criteria, identify failure modes and edge cases, annotate executions, and enforce data-quality thresholds used in model development.
- Triage recurring hardware-software faults across the robotics stack, trace service logs to isolate root causes, and document reproducible findings for engineering teams.

### Handshake AI

Apr 2026 – Jun 2026

*Contracted AI Response Evaluator*

*Remote*

- Evaluated LLM-generated code and technical responses across Python, C#/.NET, AI/ML, and software engineering for correctness, instruction following, and technical clarity.

### Castline Studio

Jan 2026 – Apr 2026

*Operator & Developer – Resin 3D Printing, Next.js, TypeScript*

*Vallejo, CA*

- Operate resin 3D-printing production for an e-commerce business generating \$2K+/month across 100+ product listings, including calibration, curing, post-processing, and tolerance control.
- Built an order-management platform with client-side STL parsing for automated geometry, volume, dimension, and print-cost calculations.

### Kaiser Permanente Regional Laboratory

Oct 2023 – Jun 2024

*Software Developer*

*Berkeley, CA*

- Developed a full-stack application using React, Express, RESTful APIs, and SQL for uploading, processing, and reviewing clinical laboratory instrument data across multiple departments.
- Added validation and error logging to a 50+ column instrument-data workflow; diagnosed faulty recordings and documented recurring failure patterns to improve pipeline reliability.

## Selected Projects

### $\pi$ 0 on a Budget | *OpenPI, LoRA, LeRobot, JAX, WSL2*

- Built a synchronized data pipeline for recording 30 FPS Logitech Brio video and joint commands from a custom 4-DOF teleoperation arm, with conversion into a LeRobot-compatible dataset.
- Implemented an RTX 4070 workflow for  $\pi$ 0-FAST fine-tuning, memory profiling, policy serving, open- and closed-loop evaluation, and a real-time dashboard that visualizes live Arduino telemetry and RTX-predicted action chunks.

### GELLO-Style Leader-Follower Teleoperation Rig | *Mechanical Assembly, Servo Control, Calibration*

- Built a four-DOF leader-follower mechanism for approximately \$30 by assembling linkages, wiring potentiometer-based joint encoders, and calibrating direct joint-space control without inverse kinematics.

### UR5e Fault Injection & Diagnostics Harness | *RTDE, URSim, Fault Injection*

- Built a control and fault-characterization harness that drives simulated robot motion and logs synchronized joint telemetry while testing connection failures, stale handles, hangs, and native crashes.

### CAN Bus Fault-Injection & Diagnostics Rig | *SocketCAN, MCP2515, Multimeter Diagnostics*

- Assembled a three-node CAN network with Arduino/MCP2515 nodes, a USB-CAN interface, and a custom wiring harness, then induced bus-off, arbitration, connector, wiring, and termination faults.
- Mapped physical failure modes to reproducible SocketCAN error states and multimeter evidence in a published diagnostic catalog.

### Fault-Tolerant Robot Telemetry Pipeline | *Kubernetes, Redis Streams, PostgreSQL*

- Designed a multi-stage telemetry pipeline for deliberate pod and network failures using durable acknowledgements, replay-safe writes, health probes, observability, and a chaos-testing harness.

Research Publications

Do Published HumanEval Rankings Survive Local Deployment?

- Reproduced a five-model HumanEval ranking on an RTX 4070 and identified a local ranking reversal; released as a technical report.

When Does INT8 Actually Accelerate Host-CPU Inference?

- Audited MobileNetV2 INT8 conversion and packaging, showing that acceleration depends on delegate and threading configuration; archived on Zenodo.

Conditioning and Directionality in Adversarial Transfer on CIFAR-10

- Evaluated directional transfer across three CNN families with multiple seeds and denominator-aware metrics; revising the manuscript following a TMLR rejection.

Not All Bad Demonstrations Are Equally Bad

- Ran 660 controlled policy fits across five demonstration failure modes, two policy families, ten seeds, and transition-matched controls; submitted to the CoRL 2026 LfC workshop.

Smoothness Ranks Skill, Not Success

- Audited published trajectory-smoothness metrics under episode-length controls and found they primarily rank operator skill rather than task success; submitted in venue-specific forms to three workshops.

Technical Skills

**Robotics & Hardware Integration:** CAN/SocketCAN, RTDE, URSim, teleoperation, real-time control, fault injection, actuator control, telemetry analysis, electrical/mechanical troubleshooting, PCB rework

**Machine Learning & Physical AI:** PyTorch, JAX, LeRobot, OpenPI, LoRA, NumPy, pandas, scikit-learn, ONNX, model evaluation

**Software, Systems & Data:** Linux, Git, Docker, ROS2, Kubernetes, Redis Streams, PostgreSQL, REST APIs, AWS

**Languages:** Python, C++, TypeScript, JavaScript, SQL, Bash

Education

Grand Canyon University

M.S. in Computer Software Engineering

May 2024 – Oct 2025

Phoenix, AZ

- Capstone: Developed an AI model-packaging CLI that converts PyTorch, TensorFlow, ONNX, and scikit-learn models into deployable Docker images or Python packages.

University of Silicon Valley

B.S. in Computer Science

May 2021 – Aug 2023

San Jose, CA

- Capstone: Developed a MetaHuman lip-sync and audio-animation pipeline in Unreal Engine for real-time playback.

Certifications

Accelerated Computing in Modern CUDA C++ (Sep 2026)

Cisco Network Support and Security (Sep 2025)

Stanford AI and Machine Learning in Healthcare (Jun 2025)

Google IT Support Professional Certificate (May 2021)

HackerRank Software Engineer Certificate (Mar 2026)

IBM Artificial Intelligence Fundamentals (Aug 2025)

Google Foundations of Project Management (Sep 2024)